LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fourth laboratory (Variant 4.1)

Timers/counters and ADC converters of the 32 bit microcontrollers

Mandatory part (6 points):

- Configure the timer/counter TIM4 so that pulse-width modulation logic can be implemented for the onboard PG13 and PG14 LEDs. Since these LEDs are not directly connected to TIM4 channels on the STM32F429I-DISC1 board, implement the task programmatically using the TIM4 update interrupt. Set PG13 to change state every 2 seconds, and set PG14 to remain on for 3.5 seconds and off for 0.5 second within a 4-second period (2 points).

Notes:

  - ✓ First, select the system frequency for the TIM4 timer/counter. Then choose the divider and AutoReload values to obtain the exact time period required by the task. Finally, choose an appropriate software time step so that the LED control matches the required period and duty cycle.
  - ✓ The purpose of the following parameters should be evaluated during the task: Prescaler, Period, interrupt period, and software LED control in the callback function.

- Write a program to measure the time interval between two presses of the user button. Implement the task for pin PA0 using the alternative function of timer/counter TIM2 in Input Capture Direct Mode. Complete the task using HAL_TIM_IC_Start_IT() and HAL_TIM_IC_CaptureCallback(). The button press must be recorded within a 12-second interval. While evaluating the captured timer value, show the LED indication given in Table 1. For time indication during the measurement, use a short PG14 blink every 1 second instead of a separate blue LED; after the second press, display the result according to the table.

  Table 1. LED indication variants.

Time window within a 12-second period	Illuminated LEDs
0 < t <= 4	Green
4 < t <= 8	Red
8 < t <= 12	Green and red

Other important functions: __HAL_TIM_GET_COUNTER(htim), __HAL_TIM_SetCounter(), HAL_TIM_IC_CaptureCallback(...).

Take switch bouncing into account either programmatically or by selecting appropriate timer input filter parameters. The measurement result may also be monitored in Debug mode (4 points).

Additional parts (4 points):

- Configure the project to use ADC1 channel 0 for sampling the analog voltage applied to pin PA0. Complete the task using HAL_ADC_PollForConversion(), and consider during configuration whether an interrupt mechanism is required. Use the following ADC1 parameters: converter resolution - 12 bit; data alignment - right; number of conversions - 1; external trigger source for conversion - regular conversion started from program code. Other parameters may be selected freely (2 points).
- Write a program that reads, in real time and without delay, the analog voltage applied to pin PA0 using HAL_ADC_PollForConversion(). When the user button is not pressed, the voltage level on PA0 is low; when the user button is pressed, the voltage rises to the supply level. Implement two indications: when the voltage on PA0 is low, the green LED is on; when the voltage on PA0 is high, both LEDs are on. Additionally, monitor the converted analog voltage value in Debug mode. Other important functions: HAL_ADC_GetValue() (2 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fourth laboratory (Variant 4.2)

Timers/counters and ADC converters of the 32 bit microcontrollers

Mandatory part (6 points):

- Configure the timer/counter TIM4 so that pulse-width modulation logic can be implemented for the onboard PG13 and PG14 LEDs. Since these LEDs are not directly connected to TIM4 channels on the STM32F429I-DISC1 board, implement the task programmatically using the TIM4 update interrupt. Set PG13 to change state every 3 seconds, and set PG14 to remain on for 4.5 seconds and off for 1.5 seconds within a 6-second period (2 points).

Notes:

  - ✓ First, select the system frequency for the TIM4 timer/counter. Then choose the divider and AutoReload values to obtain the exact time period required by the task. Finally, choose an appropriate software time step so that the LED control matches the required period and duty cycle.
  - ✓ The purpose of the following parameters should be evaluated during the task: Prescaler, Period, interrupt period, and software LED control in the callback function.

- Write a program to measure the time interval between two presses of the user button. Implement the task for pin PA0 using the alternative function of timer/counter TIM5 in Input Capture Direct Mode. Complete the task using HAL_TIM_IC_Start_IT() and TIM5_IRQHandler(void). The button press must be recorded within a 0-9 second interval. While evaluating the captured timer value, show the LED indication given in Table 1. For time indication during the measurement, use a short PG13 blink every 3 seconds instead of a separate blue LED; after the second press, display the result according to the table.

Table 1. LED indication variants.

Time window within a 9-second period	Illuminated LEDs
0 < t <= 3	Green
3 < t <= 6	Red
6 < t <= 9	Green and red

Other important functions: __HAL_TIM_GET_COUNTER(htim), __HAL_TIM_SetCounter(), TIM5_IRQHandler(...).

Take switch bouncing into account either programmatically or by selecting appropriate timer input filter parameters. The measurement result may also be monitored in Debug mode (4 points).

Additional parts (4 points):

- Configure the project to use ADC1 channel 0 for sampling the analog voltage applied to pin PA0. Complete the task using HAL_ADC_Start_IT(&hadc1) and HAL_ADC_ConvCpltCallback(), and consider during configuration whether an interrupt mechanism is required. Use the following ADC1 parameters: converter resolution - 12 bit; data alignment - right; number of conversions - 1; external trigger source for conversion - regular conversion started from program code. Other parameters may be selected freely (2 points).
- Write a program that reads the analog voltage applied to pin PA0 every 2 seconds using HAL_ADC_Start_IT(&hadc1) and HAL_ADC_ConvCpltCallback(). When the user button is not pressed, the voltage level on PA0 is low; when the user button is pressed, the voltage rises to the supply level. Implement two indications: when the voltage on PA0 is low, the red LED is on; when the voltage on PA0 is high, both LEDs are on. Other important functions: HAL_ADC_GetValue() (2 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fourth laboratory (Variant 4.3)

Timers/counters and ADC converters of the 32 bit microcontrollers

Mandatory part (6 points):

- Configure the timer/counter TIM4 so that pulse-width modulation logic can be implemented for the onboard PG13 and PG14 LEDs. Since these LEDs are not directly connected to TIM4

channels on the STM32F429I-DISC1 board, implement the task programmatically using the TIM4 update interrupt. Set PG13 to change state every 1 second, and set PG14 to remain on for 1.5 seconds and off for 0.5 second within a 2-second period (2 points).

Notes:

- ✓ First, select the system frequency for the TIM4 timer/counter. Then choose the divider and AutoReload values to obtain the exact time period required by the task. Finally, choose an appropriate software time step so that the LED control matches the required period and duty cycle.
- ✓ The purpose of the following parameters should be evaluated during the task: Prescaler, Period, interrupt period, and software LED control in the callback function.

- Write a program to measure the time interval between two presses of the user button. Implement the task for pin PA0 using the alternative function of timer/counter TIM5 in Input Capture Direct Mode. Complete the task using HAL_TIM_IC_Start_IT() and HAL_TIM_IC_CaptureCallback(). The button press must be recorded within a 0-6 second interval. While evaluating the captured timer value, show the LED indication given in Table 1. For time indication during the measurement, use a short blink of both LEDs every 2 seconds instead of a separate green LED; after the second press, display the result according to the table.

Table 1. LED indication variants.

Time window within a 6-second period	Illuminated LEDs
0 < t <= 2	Green
2 < t <= 4	Red
4 < t <= 6	Green and red

Other important functions: __HAL_TIM_GET_COUNTER(htim), __HAL_TIM_SetCounter(), HAL_TIM_IC_CaptureCallback(...).

Take switch bouncing into account either programmatically or by selecting appropriate timer input filter parameters. The measurement result may also be monitored in Debug mode (4 points).

Additional parts (4 points):

- Configure the project to use ADC1 channel 0 for sampling the analog voltage applied to pin PA0. Complete the task using HAL_ADC_Start_IT(&hadc1) and ADC_IRQHandler(void), and consider during configuration whether an interrupt mechanism is required. Start the ADC conversion programmatically from the TIM2_IRQHandler(void) interrupt function of timer/counter TIM2. Use the following ADC1 parameters: converter resolution - 12 bit; data alignment - right; number of conversions - 1; external trigger source for conversion - regular conversion started from program code. Other parameters may be selected freely (2 points).
- Write a program that reads the analog voltage applied to pin PA0 every 1 second using HAL_ADC_Start_IT(&hadc1) and ADC_IRQHandler(void). Trigger the conversion every 1 second from TIM2_IRQHandler(void). When the user button is not pressed, the voltage level on PA0 is low; when the user button is pressed, the voltage rises to the supply level. Implement two indications: when the voltage on PA0 is low, the green LED is on; when the voltage on PA0 is high, the red and green LEDs are on. Other important functions: HAL_ADC_GetValue() (2 points).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Fourth laboratory (Variant 4.4)

Timers/counters and ADC converters of the 32 bit microcontrollers

Mandatory part (6 points):

- Configure the timer/counter TIM4 so that pulse-width modulation logic can be implemented for the onboard PG13 and PG14 LEDs. Since these LEDs are not directly connected to TIM4

channels on the STM32F429I-DISC1 board, implement the task programmatically using the TIM4 update interrupt. Set PG13 to change state every 3 seconds, and set PG14 to remain on for 2 seconds and off for 1 second within a 3-second period (2 points).

Notes:

- ✓ First, select the system frequency for the TIM4 timer/counter. Then choose the divider and AutoReload values to obtain the exact time period required by the task. Finally, choose an appropriate software time step so that the LED control matches the required period and duty cycle.
- ✓ The purpose of the following parameters should be evaluated during the task: Prescaler, Period, interrupt period, and software LED control in the callback function.

- Write a program to measure the time interval between two presses of the user button. Implement the task for pin PA0 using the alternative function of timer/counter TIM2 in Input Capture Direct Mode. Complete the task using HAL_TIM_IC_Start_IT() and TIM2_IRQHandler(void). The button press must be recorded within a 0-14 second interval. While evaluating the captured timer value, show the LED indication given in Table 1. For time indication during the measurement, use a short PG14 blink every 1 second instead of a separate blue LED; after the second press, display the result according to the table.

Table 1. LED indication variants.

Time window within a 14-second period	Illuminated LEDs
0 < t <= 3	Green
3 < t <= 10	Red
10 < t <= 14	Green and red

Other important functions: __HAL_TIM_GET_COUNTER(htim), __HAL_TIM_SetCounter(), TIM2_IRQHandler(...).

Take switch bouncing into account either programmatically or by selecting appropriate timer input filter parameters. The measurement result may also be monitored in Debug mode (4 points).

Additional parts (4 points):

- Configure the project to use ADC1 channel 0 for sampling the analog voltage applied to pin PA0. Complete the task using HAL_ADC_PollForConversion(), and consider during configuration whether an interrupt mechanism is required. Use the following ADC1 parameters: converter resolution - 12 bit; data alignment - right; number of conversions - 1; external trigger source for conversion - regular conversion started from program code. Other parameters may be selected freely (2 points).
- Write a program that reads, in real time and without delay, the analog voltage applied to pin PA0 using HAL_ADC_PollForConversion(). When the user button is not pressed, the voltage level on PA0 is low; when the user button is pressed, the voltage rises to the supply level. Implement two indications: when the voltage on PA0 is low, the red LED is on; when the voltage on PA0 is high, the red and green LEDs are on. Additionally, monitor the converted analog voltage value in Debug mode. Other important functions: HAL_ADC_GetValue() (2 points).